A Comparative Study of Machine Learning Techniques for Cyber-Physical Attack Detection

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***Abstract*—Cyber-physical systems (CPS) require attack-detection mechanisms that remain accurate, timely, and reliable under strict operational constraints. This paper presents a comparative study of machine learning techniques for detecting malicious activity in CPS network traffic, with emphasis on Modbus/TCP environments. We examine unsupervised anomaly detection, classical supervised classification, sequence-aware deep learning, and an exploratory reinforcement-learning formulation. In addition, we analyze the effect of feature selection, preprocessing, and model configuration on detection performance, and use SHAP analysis to interpret the most influential traffic features. The results show that supervised models substantially outperform unsupervised baselines, while a recurrent neural network achieves the strongest predictive performance by exploiting temporal context. The study also highlights practical trade-offs among accuracy, generalization, interpretability, and computational complexity, providing guidance on the suitability of different learning paradigms for real-time CPS security.**

***Index Terms*—Cyber-physical systems, attack detection, machine learning, anomaly detection, deep learning, classification.**

I. Introduction

Cyber-physical systems (CPS) underpin critical infrastructure in sectors such as energy, water, manufacturing, and transportation. These environments tightly couple physical processes with control logic executed on programmable logic controllers (PLCs) and rely on industrial communication protocols to coordinate sensors, actuators, and supervisory control systems. Historically, operational technology (OT) networks were largely isolated from information technology (IT) infrastructures; however, increasing IT/OT convergence for remote monitoring, analytics, and centralized control has significantly expanded system connectivity and complexity [1].

Although this convergence improves visibility and operational efficiency, it also exposes CPS to a broader range of cyber-induced disruptions. Network anomalies such as latency spikes, congestion, packet manipulation, or unauthorized access can interfere with time-sensitive control traffic and, in turn, affect physical processes. Detecting such events is particularly challenging because their manifestations often resemble benign operational variability or process disturbances. Accurate and timely detection is therefore essential to preserving safety, reliability, and service continuity.

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Machine learning (ML) has emerged as a promising approach for CPS attack detection because it can model complex patterns in high-dimensional traffic data. Prior work has explored supervised classification, unsupervised anomaly detection, and deep learning models; however, these approaches are often evaluated in isolation, making it difficult to understand their relative strengths, limitations, and deployment implications in CPS environments. As a result, the trade-offs among detection performance, generalization, interpretability, and computational cost remain insufficiently characterized.

This paper presents a comparative study of machine learning techniques for attack detection in CPS network traffic, with a focus on Modbus/TCP communication. We compare multiple paradigms, including unsupervised anomaly detection methods (e.g., Isolation Forest and Local Outlier Factor), classical supervised classifiers (e.g., Random Forest), sequence-aware deep learning, and an exploratory reinforcement-learning formulation. We further investigate how feature selection, preprocessing, and model configuration influence performance, and we use feature-importance analysis to interpret model behavior.

The main contributions of this work are as follows:

- A comparative evaluation of unsupervised, supervised, deep learning, and exploratory reinforcement-learning approaches for CPS attack detection.
- An analysis of how feature engineering and model design affect detection accuracy, robustness, and interpretability.
- A discussion of the trade-offs among predictive performance, temporal modeling, and practical deployment considerations in CPS environments.

The remainder of this paper is organized as follows. Section II summarizes CPS fundamentals and the role of ML in attack detection. Section III reviews related work. Section IV describes the workflow, datasets, and feature set. Section V presents the experimental results. Finally, Section VI concludes the paper and outlines future directions.

II. Background

A. Cyber-Physical Systems

CPSs with Supervisory Control and Data Acquisition (SCADA) systems are widely deployed in critical infrastructure domains such as manufacturing, energy, and water treatment. These systems typically follow a hierarchical architecture that separates supervisory functions from physical process

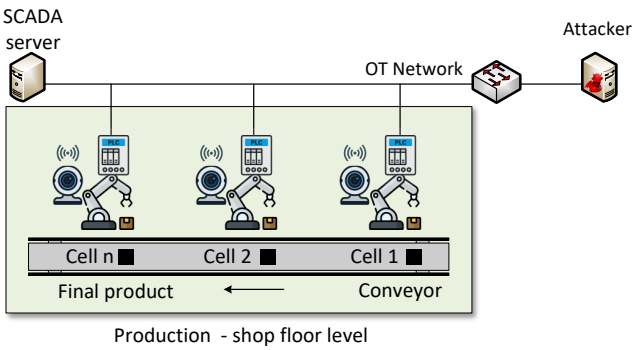

Fig. 1. Representative CPS architecture for network-based attack detection.

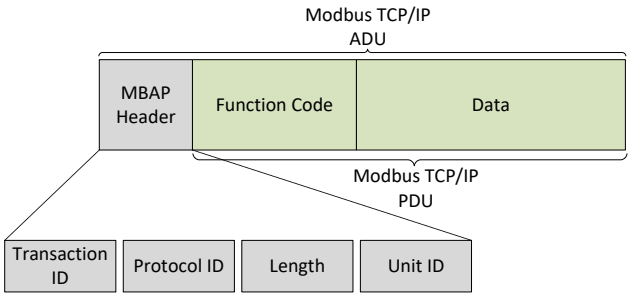

Fig. 2. Structure of a Modbus/TCP packet.

control across multiple layers. At the supervisory layer, the SCADA server supports monitoring, visualization, and high-level coordination. The control layer consists of programmable logic controllers (PLCs), which execute real-time control logic and interact with field devices using industrial communication protocols such as Modbus/TCP.

The adoption of Industry 4.0 paradigms, including real-time analytics and remote supervision, has increased the reliance on IP-based networking within OT environments [2]. As illustrated in Fig. 1, SCADA servers often operate as aggregation points for high-volume sensor data and control traffic generated by multiple PLCs across the production floor. This connectivity improves operational flexibility, but it also increases the attack surface of CPS networks.

Consequently, there is a growing need for effective intrusion and anomaly detection mechanisms. ML-based approaches are promising because they can analyze communication patterns and identify deviations from expected behavior. By inspecting the packets exchanged between SCADA servers and PLCs, ML models can help detect cyber-attacks or misconfigurations in real time, thereby improving the security and resilience of CPS environments.

B. Modbus TCP/IP

Modbus TCP/IP is a communication protocol designed to facilitate data exchange between industrial devices, including controllers and sensors, over Ethernet networks. As shown in Fig. 2, a Modbus TCP/IP Application Data Unit (ADU) consists of two main components: the MBAP (Modbus Application Protocol) header and the Protocol Data Unit (PDU). The MBAP header includes the Transaction ID, Protocol ID, Length, and Unit ID. The Transaction ID uniquely associates a request with its corresponding response, enabling message tracking within a communication session, while the Unit ID identifies the target PLC. Together, the Transaction ID and Unit ID define a specific communication flow between a client and a given PLC. The Function Code specifies the requested operation, such as reading or writing data, and the Data field carries the corresponding values in byte format [3].

C. Impact of Cyber Attacks on Cyber-Physical Systems

The convergence of IT and OT networks has significantly increased the attack surface of industrial CPSs. Unlike traditional IT systems, attacks on CPS can directly affect physical processes, leading to safety hazards, equipment damage, and operational disruption. Recent incidents targeting critical infrastructure such as power grids and manufacturing systems illustrate the severity of these threats [4].

Common attack vectors in CPS include false-data injection, command manipulation, denial-of-service (DoS), and replay attacks, all of which can degrade control performance or destabilize process behavior. These risks are compounded by the widespread use of legacy industrial protocols such as Modbus/TCP, which provide limited native security.

To mitigate these challenges, ML-based techniques have gained traction for detecting anomalies in network traffic and system behavior [5]. By learning normal operational patterns and identifying deviations, ML approaches can support early attack detection and improve the resilience of CPS environments.

III. Related Work

Prior research on CPS and ICS security spans supervised classification, anomaly detection, deep learning, and deployment-oriented optimizations. Broad surveys highlight the distinctive constraints of industrial environments, including protocol heterogeneity, strict timing requirements, and the tight coupling between cyber events and physical processes [4], [5].

In the supervised setting, Jan *et al.* combine distributed processing with auto-encoders and SVMs for sensor-fault detection, while Doshi *et al.* demonstrate the effectiveness of gradient-boosted trees for anomaly detection in industrial robotic systems [6], [7]. Random-Forest-based diagnosis has also shown strong performance in cyber-physical contexts because of its robustness to heterogeneous features and modest preprocessing requirements [8]. Collectively, these studies indicate that labeled-data methods can achieve high accuracy, but they are typically evaluated in task-specific settings and do not always provide a direct comparison across different learning paradigms.

A complementary research direction emphasizes deployment efficiency. Elizalde *et al.* show that attack-detection inference can be accelerated in industrial networks by offloading models to SmartNICs, underscoring the importance

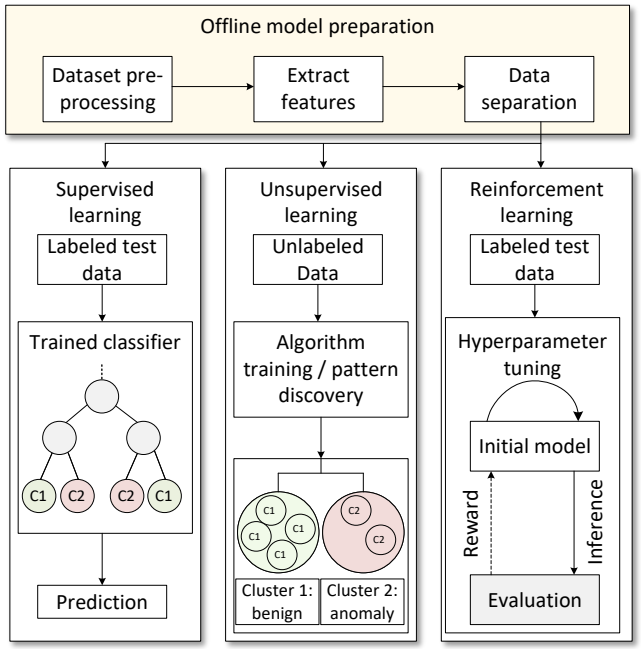

Fig. 3. Offline training and online inference workflow.

of latency-aware implementation in OT environments [9]. This perspective is highly relevant to CPS security, where detection quality and inference overhead must be considered jointly.

Despite this progress, comparatively fewer studies provide a unified assessment of unsupervised anomaly detection, classical supervised learning, sequence-based deep models, and reward-driven learning under a common traffic-analysis pipeline. This paper addresses that gap by examining these paradigms side by side and by analyzing the contribution of feature design and temporal context to attack detection performance.

IV. Methodology

The proposed framework follows a two-stage workflow that separates offline model preparation from online inference. As illustrated in Fig. 3, raw CPS network traces are first processed to extract flow-level statistical features from Modbus traffic. These features are then used to train and evaluate three learning paradigms: supervised learning, unsupervised learning, and reinforcement learning.

In the supervised setting, labeled data are used to train classifiers that distinguish benign from attack traffic. In the unsupervised setting, models attempt to identify anomalous behavior without access to attack labels during training. The reinforcement-learning formulation models detection as a sequential decision problem and optimizes a policy from reward feedback.

After training, each model is applied to unseen traffic to emulate an online inference stage. Incoming flows are processed in real time, transformed into the selected feature representation, and passed to the trained detector to classify system behavior and generate alerts.

TABLE I
Zenodo ICS vs. DataSense: shared characteristics.

Characteristic	**Zenodo ICS** [10]	**DataSense** [11]
Testbed	Bottle-filling CPS	IIoT lab (40 devices)
Modalities	Sensors, logs, Modbus PCAP	Sensors, network, logs
Faults	5 types	—
Attacks	4 scenarios	7 categories (∼50 scen.)
Focus	Fault vs. cyber	Intrusion detection
Packets	4.0 M	0.26 M / 1.82 B
Logs	1.66 M	72.5 k / 1.63 M
Sensor reads	10.3 k	—

Packets/logs cells: benign / attack where applicable.

A. Datasets

This study considers the public Zenodo ICS Dataset [10] and the recent large-scale IIoT dataset DataSense [11]. The two benchmarks are complementary: the Zenodo dataset captures the interplay between physical faults and cyber events in a compact CPS, whereas DataSense emphasizes cyberattack diversity in a heterogeneous IIoT environment.

The Zenodo ICS Dataset was generated on an extended bottle-filling CPS with two PLCs and two HMIs, and provides three timestamp-aligned data sources: PLC sensor readings, PLC/HMI system logs, and Modbus/TCP PCAPs [10]. It covers normal operation, five injected faults (sensor drift, tank leak, PLC-delay overheating, valve malfunction, and memory corruption), and four cyberattacks (reconnaissance, DoS, MitM, and replay). The traffic and telemetry portion considered in this study contains 4015333 Modbus/TCP packets, 1663672 Syslog messages, and 10340 sensor readings.

DataSense is collected from a heterogeneous IIoT laboratory that combines industrial sensors, IoT devices, network equipment, edge/cloud components, and attacker nodes, while likewise synchronizing sensor, network, and log streams [11]. Its focus is attack-oriented rather than fault-oriented: it provides benign traffic together with seven attack categories (reconnaissance, DoS, DDoS, web, MitM, brute-force, and malware) across roughly fifty scenarios.

Table I summarizes the two datasets. Both support multimodal anomaly analysis, but only the Zenodo dataset explicitly models physical faults, whereas DataSense offers broader cyberattack diversity and a substantially larger attack-traffic scale.

B. Feature Selection and Importance

To characterize CPS network behavior, a pool of flow-level statistical features was extracted from Modbus traffic. These features include packet-size statistics (minimum, maximum, and mean), inter-arrival-time (IAT) statistics, and protocol-specific indicators such as request and response counts. Together, they capture both timing behavior and protocol semantics that are relevant to distinguishing benign activity from attack traffic.

Fig. 4 presents the SHAP-based feature-importance analysis. Features such as *modbus req*, *pkt size mean*, and *iat mean*

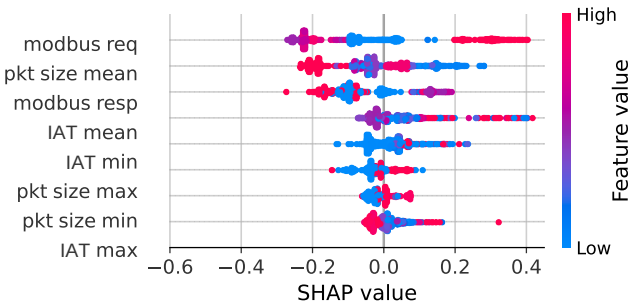

Fig. 4. SHAP summary of feature influence.

TABLE II
PERFORMANCE OF CLASSICAL SUPERVISED MODELS. BEST VALUE PER COLUMN IN BOLD.

Model	Acc.	Prec.	Recall	F1	MCC	AUC
Random Forest	**0.9525**	**0.9613**	**0.9511**	**0.9561**	**0.9044**	**0.9864**
SVM (RBF)	0.8609	0.9377	0.7976	0.8620	0.7333	0.9227
MLP	0.9297	0.9554	0.9136	0.9340	0.8599	0.9734

exhibit the strongest influence on model predictions, while the remaining features provide complementary context. This result suggests that both traffic dynamics and protocol-level behavior are important for CPS attack detection.

V. EVALUATION AND RESULTS

This section compares the proposed feature pipeline across four complementary settings: classical supervised learning, sequence-aware deep learning, unsupervised anomaly detection, and reinforcement learning. The first two settings establish label-driven baselines, the third quantifies performance when labels are unavailable during training, and the fourth explores reward-driven optimization for sequential decision making.

A. Supervised Learning

Supervised models were trained on labeled flow records to classify traffic as benign or attack. Two categories of supervised models are considered: classical machine learning algorithms that operate on tabular features under an i.i.d. assumption, and a deep learning model that captures temporal dependencies in sequential data.

1) Classical ML Algorithms: Three classical classifiers were evaluated: Random Forest (RF), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP). These models operate on individual flow records and assume independence between samples. The RF uses 400 trees with balanced subsampling, the SVM uses an RBF kernel with class-balanced weighting, and the MLP consists of two hidden layers with 64 and 32 neurons.

Table II summarizes the results, while Fig. 5 shows the corresponding ROC curves.

Discussion. The Random Forest achieves the strongest overall performance among the classical supervised models. Its ensemble structure captures nonlinear relationships effectively and handles heterogeneous feature scales without explicit normalization. The MLP performs competitively but remains slightly below RF, while the SVM exhibits lower recall, suggesting greater difficulty in modeling the decision boundary induced by this feature space.

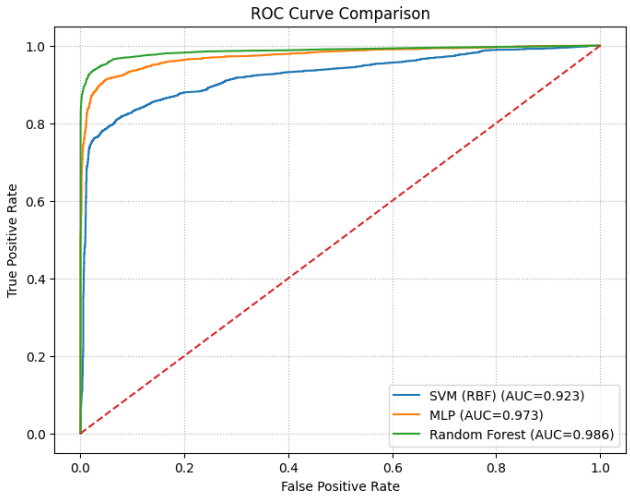

Fig. 5. ROC curves for classical supervised models.

TABLE III
DEEP LEARNING (RNN) PERFORMANCE ON SEQUENCE DATA. BEST VALUE PER COLUMN IN BOLD.

Model	Acc.	Prec.	Recall	F1	MCC	AUC
RNN	**0.9844**	**0.9889**	**0.9821**	**0.9855**	**0.9686**	**0.9983**

2) Recurrent Neural Network: To capture temporal dependencies in network traffic, a recurrent neural network (RNN) was employed. Unlike classical models that treat each flow independently, the RNN processes sequences of consecutive flows and can therefore learn temporal patterns associated with sustained attack behavior such as scanning or bursty denial-of-service activity.

Sequences of length 10 were constructed from the selected feature subset, yielding 29,688 samples of shape $(10, 6)$. The model consists of a single recurrent layer with 64 hidden units followed by a linear classification layer. Training was performed using the Adam optimizer with a learning rate of 10^{-3} and binary cross-entropy loss.

Table III reports the performance of the RNN model, and Fig. 6 presents the ROC curve.

Discussion. The RNN yields the best performance among the quantitatively evaluated supervised models. By leveraging temporal context, it improves substantially over the strongest classical baseline (e.g., MCC of 0.9686 versus 0.9044 for RF). This result indicates that sequence-aware modeling is particularly valuable in CPS and IIoT environments, where attack behavior often unfolds over time rather than within isolated flow instances.

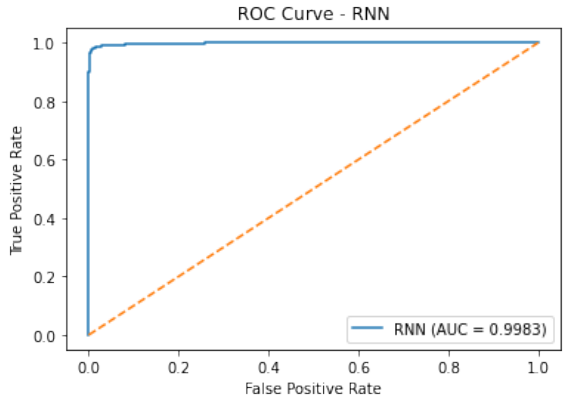

Fig. 6. ROC curve for the RNN model.

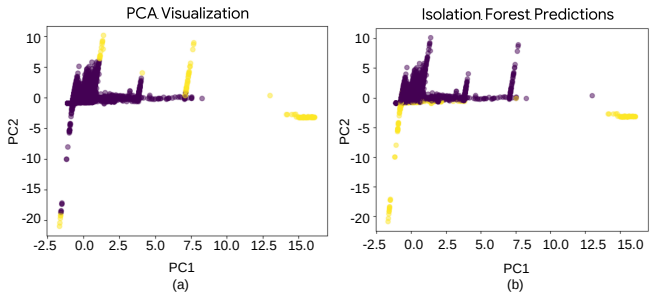

Fig. 7. PCA projection and Isolation Forest predictions.

B. Unsupervised Anomaly Detection

To quantify how well attacks can be flagged without access to labels during training, three unsupervised anomaly-detection methods were applied: Isolation Forest, DBSCAN, and Local Outlier Factor (LOF). For this experiment, the dataset was rebalanced so that attacks constitute approximately $5\,\%$ of the population, modeling a more realistic deployment scenario in which attacks are rare events. Features were standardized and, for the distance-based methods, projected onto two principal components (Fig. 7).

Table IV reports the precision, recall, F1, and MCC of each method when the anomaly flag is interpreted as a binary attack prediction. Accuracy and AUC are omitted because the class prior is heavily skewed and these methods do not produce calibrated probabilistic scores. The last column shows the percentage of points flagged as anomalous in the rebalanced set.

Discussion. All three unsupervised methods perform substantially worse than the supervised baselines, which is expected because they must rely on geometric or density assumptions rather than class labels. Isolation Forest is the strongest of the three, likely because its randomized partitioning is better suited to the high-variance tail that contains DDoS-like attack behavior. DBSCAN exhibits extremely low recall (0.013), indicating that under the selected hyperparameters most attack points are absorbed into the dense benign cluster rather than being identified as noise. LOF flags a similar fraction of points as

TABLE IV
UNSUPERVISED ANOMALY-DETECTION RESULTS. BEST VALUE PER COLUMN IN BOLD.

Model	Prec.	Recall	F1	MCC	Flagged (%)
IF	**0.604**	**0.114**	**0.192**	**0.248**	1%
DBSCAN	0.345	0.013	0.025	0.059	0.2%
LOF	0.407	0.078	0.130	0.160	1%
Attack set	-	-	-	-	5.26%

Fig. 8. Training reward of the DQN agent across episodes.

Isolation Forest but achieves lower precision because many of its outliers correspond to benign edge cases. The visualization in Fig. 7 illustrates the core difficulty: a substantial portion of the attack class remains embedded in the benign manifold, which limits the effectiveness of purely density- or isolation-based methods.

C. Reinforcement Learning

The same selected six-feature representation was used to cast attack detection as a sequential decision problem in which an agent learns from reward feedback rather than from a directly optimized supervised loss. Unlike the unsupervised setting, however, this formulation still depends on ground-truth labels through the reward function; it should therefore be interpreted as an alternative training paradigm rather than as a label-free detector.

A custom environment presents one flow record at a time as the state, and the agent's two actions correspond to the predicted class (benign or attack). A correct prediction yields a reward of $+1$ and an incorrect prediction yields -1; an episode terminates after all records in the training set have been processed. The agent is implemented as a Deep Q-Network (DQN) with two hidden layers of 64 and 32 units, trained under an ϵ-greedy policy ($\epsilon : 1.0 \rightarrow 0.05$, decay 0.995), discount factor $\gamma = 0.95$, and a 10,000-transition replay buffer. Training proceeds for 20 episodes, and the target network is synchronized every 5 episodes.

As shown in Fig. 8, the episodic reward rises from approximately 126 in episode 1 to a peak of 2,262 in episode 19, indicating that the agent learns a progressively better decision

policy over training. Because this experiment is reported through training dynamics rather than hold-out classification metrics, it should be viewed as preliminary evidence of feasibility rather than as a definitive quantitative comparison against the supervised baselines.

VI. Conclusion

This comparative study yields three main observations. First, unsupervised anomaly-detection methods perform substantially worse than supervised alternatives on the considered traffic features, which reinforces the value of labeled data whenever representative attack examples can be collected or synthesized. Second, among classical tabular models, Random Forest provides a strong and practical baseline, combining high predictive accuracy with limited preprocessing requirements. Third, the RNN achieves the best predictive performance among the quantitatively evaluated models, demonstrating that temporal context is highly informative for CPS attack detection.

The exploratory DQN experiment suggests that reward-driven learning is feasible in this setting, but a more rigorous assessment will require hold-out performance metrics and a deployment-oriented comparison with the supervised models. Future work will extend the analysis to cross-dataset generalization, multimodal fusion of traffic and process variables, and real-time deployment constraints in operational CPS environments.

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